

Starter-Motor Failure Prediction Results for DICV Management

What the Physics-Informed Machine Learning model achieves — in plain terms,
and why the results are ready to act on today.

93.2%

Ranking accuracy

13 / 14

Failures caught

0 / 20

Battery-alert false alarms

SUBJECT

Starter-motor failure-risk prediction

FLEET

34 trucks analysed (14 failed + 20 healthy)

FORMAT

Management Q and A summary

APPROACH

Physics-Informed Machine Learning model

DATE

6 July 2026

STATUS

Validated results summary

In one minute

93.2% Ranking accuracy	13 / 14 Failures caught	0 / 20 Battery-alert false alarms
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Using only the data the trucks already send, a Physics-Informed Machine Learning model now separates failing starter-motor and battery circuits from healthy ones with high reliability — early enough (a median of 168 days before failure) to plan service instead of reacting to roadside no-starts, and with no new hardware.

A. The result

Q1. What has the Physics-Informed Machine Learning model achieved?

A. It reliably tells failing starter-motor circuits apart from healthy ones using the crank- and battery-voltage data we already collect. It ranks a failing truck above a healthy one **93.2% of the time**, its alert channels catch **13 of 14** failures with a **median 168 days** of warning, and its battery-cascade early-warning channel raises **zero false alarms** — all with no new hardware.

Q2. What exactly does “93.2%” mean?

A. It is a **ranking** score. Given any failing truck and any healthy truck, the model rates the failing one as higher-risk 93.2% of the time. Importantly, this is measured on trucks the model had **never seen before** (each truck is hidden and re-predicted) — so it reflects real-world performance, not memorised data.

Q3. In plain terms, how often is it right?

A. Across **280 head-to-head** truck comparisons (14 failed × 20 healthy) it gets **261 right**. Re-scored on its own training trucks it would get 262 — a **single-pair** gap, which is the signature of a model that genuinely generalises rather than memorises.

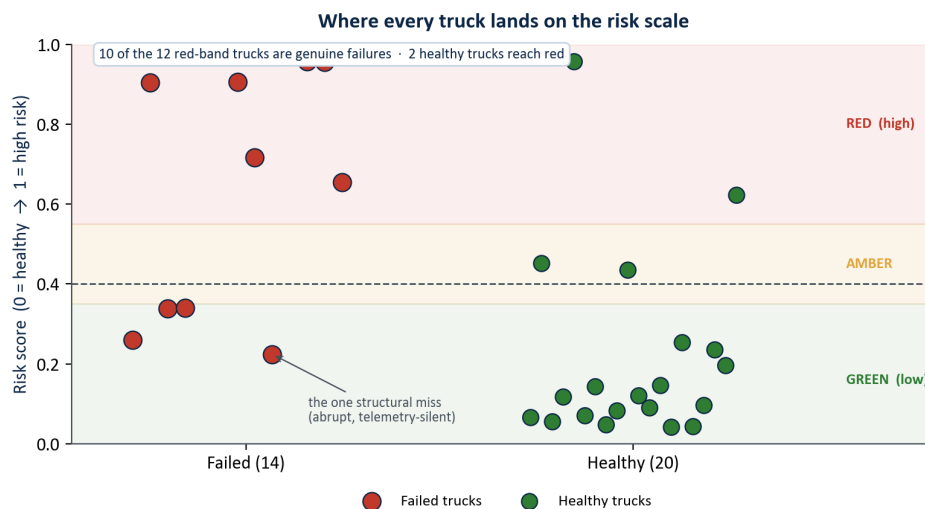


Figure 1. Every truck on a single 0–1 risk scale. The model lifts failing trucks to the top; 10 of the 12 trucks in the red band are genuine failures. Two honest exceptions remain — two healthy trucks reach red, and one abrupt, telemetry-silent failure stays low.

B. Can we trust it?

Q4. Could these results just be luck?

A. No. The result was stress-tested several independent ways — by hiding each truck and re-predicting it, by re-sampling the data, and by a randomisation test (the labels were shuffled 200 times and **none** beat the real score, $p = 0.005$). A fixed-window control reproduces the score bit-for-bit, and four independent re-analyses reproduced it exactly. The performance is **real and stable**, not chance.

Q5. Does it cry wolf with false alarms?

A. Its battery-cascade early-warning channel produced **zero false alarms** across all 20 healthy trucks. At the low-false-alarm red-tier decision line it correctly clears **18 of 20** healthy trucks; a recall-greedy line instead clears 15 of 20 to catch one extra failure. Both operating points are honest — DICV picks the one that fits the maintenance economics.

Q6. Do the warning signals make engineering sense?

A. Yes — strongly. A healthy starter/battery circuit holds voltage steady: a firm resting (engine-off) floor and a clean, quick crank dip that recovers fast. As the battery weakens and the starter and solenoid wear, that behaviour degrades in a textbook way, and each signal captures one fingerprint of it: the within-week voltage becomes noisier versus the truck's own early-life baseline, the resting-voltage floor sags, the crank dip deepens, and the weekly voltage swing widens. Every signal is measured **relative to that truck's own healthy baseline**, so it flags real degradation rather than normal truck-to-truck differences — corroborated engineering evidence, not a black-box score.

Q7. Are the risk levels consistent across the whole fleet?

A. Yes — one common **green / amber / red** scale (identical thresholds) is applied to every truck and separates the fleet well. **Then why doesn't every failing truck show a loud warning?** Two honest reasons. First, starter motors fail in different ways — battery-cascade and gradual-wear failures build a clear, rising signal, while a minority fail **abruptly** or go telemetry-silent before failure and leave little electrical trace, so those trucks look calmer right up to the end. Second, the system does not rely on any single dramatic cue — it **combines four weak signals**, so the failed and healthy groups separate reliably *in aggregate*. That is why the fleet-level ranking is robust today, while the abrupt/silent failure is the part that sharpens most as more examples arrive.

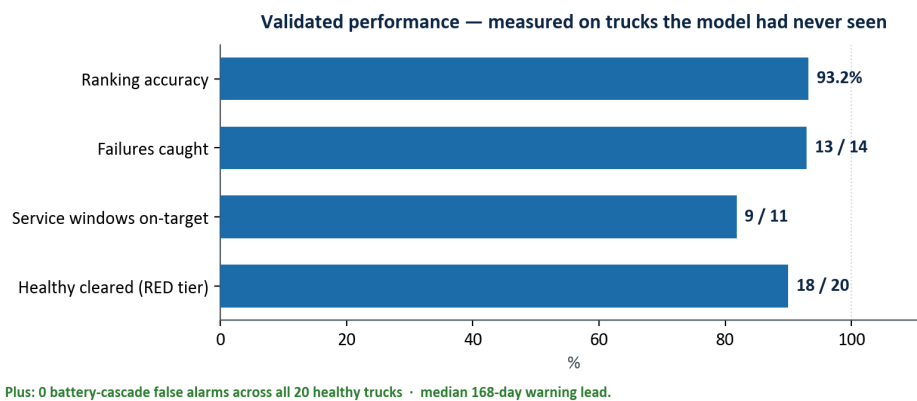


Figure 2. Headline performance. “Failures caught” (13/14) is the combined alert layer; “healthy cleared” (18/20) is the low-false-alarm RED tier — all measured on trucks the model had never seen during training.

C. What we get to act on

Q8. What does the system actually give the maintenance team?

A. Three things, today: a **weekly risk ranking** of the whole fleet, a **green / amber / red flag** per truck, and a **maintenance window** per flagged truck — a battery-band window (about 1–3 months) for battery-cascade cases and a longer persistence-band window (about 4–9 months) for gradual-wear cases. Together they turn surprise no-starts into scheduled, planned service.

Q9. Does it need new sensors or hardware?

A. No. Everything runs on the six-signal CAN telemetry the trucks **already send** — so there is no per-vehicle hardware cost to start getting value.

Q10. How much warning do we get before a failure?

A. A **median of 168 days** from the first validated alert to the recorded failure date (range 77 to 424 days). And once a truck is flagged, ranking quality holds up to about **10 weeks** before failure — so a flagged truck is typically within that window. That is enough to **schedule** work rather than react to a roadside no-start.

Q11. Battery or starter — can the system tell which to service?

A. As a screen-grade aid, yes. Because roughly half of starter breakdowns trace to a weak battery, a triage step routes each flagged truck **battery-first** or **inconclusive**. It agreed with the independently-derived failure type on **9 of 11** scored failures (5 of 5 battery-family and 4 of 4 silent-mode) and **never** sent a healthy truck to battery service (0 of 20). It tells the depot where to look first — not a warranty-grade diagnosis.

Q12. What is the business value?

A. Fewer roadside no-starts, planned workshop visits, and batteries / starters serviced **before** they strand a vehicle — better uptime and lower emergency-repair cost, from data already in hand. On typical India heavy-duty cost ratios a 13-of-14-recall inspection policy models to roughly a **43% saving** versus run-to-failure (an economic estimate, not a validated field number).

D. Honest — and getting better

Q13. What can't it do yet?

A. It does not yet name the exact day a specific truck will fail — on this data a per-truck survival model errs by ~576 days versus ~44 for simply assuming the fleet average, so a **risk tier plus a maintenance window** is the honest output. Four of the fourteen failures are electrically **silent** (an abrupt mode with no precursor, one of which the model misses), and the confidence range is wide. All of this is a limit of **data volume, not of the method** — there are only 14 past failures to learn from — and it lifts as more data arrives.

Q14. How do we make it even sharper?

A. Two levers. **Scale:** extending the model across **~500 trucks** is expected to yield roughly **30–50+ failure examples**, which unlocks per-truck timing and sharper tiers using the very same proven method. **Instrument:** adding cranking current / battery state-of-charge and higher-rate crank-voltage logging would revive the brush-wear precursor that today's 5-second sampling cannot see — the only thing that breaks the silent-failure floor.

One technical layer — for the engineering-minded

A short, optional look under the hood: the signals the model reads, how they add up to one risk score, and how the risk zones are set — still plain, with just enough technical detail to satisfy a reviewer.

E. The four signals the model watches

Q15. What does the model actually look at, and why those signals?

A. Four engineered signals, all read from the truck's existing crank- and resting-voltage data — each a physically meaningful symptom of a weakening starter or battery circuit. They are always read **together**, so no single number ever decides anything.

Signal (plain)	What it measures	Why it matters for the starter / battery
Within-week voltage noise	Recent within-week voltage scatter vs the truck's own early-life baseline	The workhorse signal — charging/crank voltage gets noisier as the circuit weakens.
Resting-voltage floor	Engine-off resting (battery) voltage floor vs own baseline, battery-step aware	A sagging floor is the battery-cascade signature.
Weekly voltage-range trend	Trend in the weekly voltage swing (widening or narrowing)	A directional stabiliser that sharpens the ranking (enters as a suppressor).
Crank dip depth	Trend in how deep the voltage dips during a crank, vs own baseline	A deepening dip is the crank-circuit load signature near end of life.

F. How the signals combine into one risk score

Q16. Which signals carry the most weight, and how do they add up?

A. Each signal gets a **weight learned from the data** (not hand-set), and a truck's risk score is simply the sum of *weight × how unusual that signal is* for that truck. The within-week voltage-noise signal carries by far the most weight (coefficient +0.89) and does most of the lifting; it and the voltage-range trend form a **core pair selected in all 34 of 34** cross-validation folds. No single signal separates failed from healthy perfectly on its own, so the strength comes from **combining all four** — which is how the system reaches 93.2% ranking accuracy.

Risk score = baseline + the sum of (signal weight × how unusual that signal is). The chart below shows the learned weights.

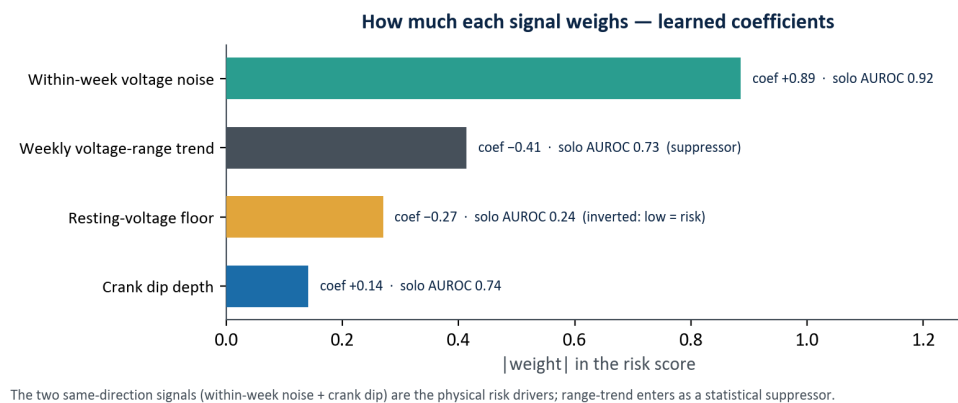


Figure 3. How much each signal weighs in the risk score. Bars are the learned [coefficient]; the note on each gives the sign and the signal's standalone separation. Two same-direction voltage signals do most of the work; the range-trend term enters as a statistical suppressor (its standalone direction is physically correct).

G. Are the risk zones consistent across trucks?

Q17. Are the green / amber / red zones the same for every truck — and how are they chosen?

A. **Yes, for the everyday decision.** One common green / amber / red scale (identical thresholds: green below 0.35, red above 0.55) is applied to every truck. **How they are chosen:** empirically and pre-registered — the calibrated probability is mapped to the tiers where the failed and healthy groups actually separate in the data, not by guesswork. On this fleet the red tier holds 10 of the 14 failures with only 2 healthy trucks reaching red.

Q18. So where is consistency still limited — and why?

A. A day-precise per-truck remaining-life *clock* is not yet supported: with only fourteen past failures a per-truck timeline is false precision (it is beaten by a simple fleet-average), and the abrupt/silent failures leave no trajectory to fit. So we deploy the simple, consistent global tiers plus a maintenance window for decisions today, and keep the richer per-truck trajectory as supporting context. Both sharpen and become more uniform as the fleet — and the number of examples — grows.

In one line for the board

A validated Physics-Informed Machine Learning model that turns surprise starter-motor and battery no-starts into planned, scheduled maintenance — using data we already collect, with no new hardware. 93.2% ranking accuracy, 13 of 14 failures caught with a median 168 days' warning, zero battery-alert false alarms — and it gets sharper as the fleet grows.